

Revision Notes for Class 8 Science Chapter 2 The Invisible Living World

We are surrounded by a world that mostly remains hidden from our ordinary eyesight. All around us, tiny organisms and structures exist that we can only discover with the help of tools like magnifying glasses and microscopes. The idea of a lens, a simple curved piece of glass, changed how humans observe the world. With improved lenses over time, we uncovered a realm of living beings too small to be seen directly. Among these are bacteria, protozoa, fungi, and algae—collectively called microorganisms or microbes.

These organisms make up an incredible variety in nature. Some living beings are big, like animals and trees, but many are very tiny. All living beings, whether plants or animals, are called organisms, and every organism, no matter its size, is made from building blocks called cells.

Microscopes Open a Hidden World

The invention of the microscope in the 17th century was a turning point in science. Robert Hooke, using his early microscope, looked at cork and saw tiny compartments, which he named 'cells'. Around this time, Antonie van Leeuwenhoek made stronger lenses and was the first to observe living microbes, like bacteria and blood cells. Today, we know these microscopic living forms play vital roles in our lives.

Structure and Types of Cells

Cells are the basic structural and functional units of life. You can observe plant cells by looking at onion peels under a microscope. Animal cells, like the ones found in the lining of your cheek, look different compared to plant cells. Most cells have three main parts: the cell membrane (outer boundary), the cytoplasm (jelly-like substance inside), and the nucleus (controls all activities of the cell). Plant cells also have an extra layer called the cell wall for added strength, and contain plastids such as chloroplasts.

Cells show much variety. For instance, muscle cells are spindle-shaped to help contraction, while nerve cells have long branches for quickly transmitting messages. The shape of a cell often matches its function, from flat cheek cells forming linings to long, tubular plant cells transporting water.

Levels of Organisation in Living Beings

In multicellular organisms, the body is organised in levels. The smallest unit is the cell. Similar cells form tissue. Different tissues make up organs. Groups of organs work together in organ systems, which together make up the organism. The whole process starts from a single cell in most animals and plants. For example, the yolk of an ostrich egg is the largest known cell.

Microorganisms: Tiny Life Forms Everywhere

Many organisms consist of only a single cell. These are called unicellular organisms (like bacteria and Amoeba). Microorganisms can be unicellular or multicellular (some fungi and algae). They live in soil, water, air, food, and even inside our bodies. Observing a drop of pond water under a microscope will reveal tiny protozoa and algae. Examining soil suspension shows bacteria and different types of moulds and fungi. Viruses, even smaller than bacteria, can only reproduce inside living cells.

Our Connection to Microbes

Microorganisms affect us in everyday life. They cause food to rot and spoil, as seen when fruits are left out and develop mouldy growths. But, foods rich in salt or sugar, like pickles and murabbas, are protected as these substances act as preservatives and prevent microbial activity. Microbes can also live in extreme places like hot springs, snow, and even inside the human digestive system.

Microbes Help Clean the Environment

Some microbes are nature's cleaners. If you bury vegetable peels into soil, after a couple of weeks, they get turned into dark, rich manure because of the action of bacteria and fungi. These decomposers break down waste matter and recycle nutrients, keeping the

environment clean. Microbes are also responsible for producing biogas—a useful fuel—by breaking down waste in places where there is no oxygen.

Indian scientific heritage has acknowledged these microbes as “Krimi” in ancient texts. Today, modern science also utilises them, for example, Dr. Ananda Mohan Chakrabarty created bacteria for cleaning oil spills, showing the relevance of microbes in solving real problems.

Microorganisms in Food Production

Microbes play a key role in preparing many foods. When yeast is added to dough, it makes it rise and become fluffy due to the production of carbon dioxide. This is used in making bread, cakes, and traditional Indian foods like idlis and bhaturas. Bacteria called Lactobacillus convert milk into curd by fermenting milk sugar to lactic acid, especially in warm conditions. Some bacteria in the nodules of legume roots, like beans, trap nitrogen from the air, which adds fertility to the soil.

Microalgae: Tiny Oxygen Factories

Microalgae are very small plant-like organisms living in water, soil, or even on trees. They produce more than half of the earth’s oxygen and serve as food for aquatic animals. Some, like Spirulina and Chlorella, are highly nutritious for both animals and humans. Due to environmental changes and pollution, conserving microalgae has become important for maintaining the oxygen balance and supporting livelihoods.

Cells: The Universal Building Blocks

Cells form the basis of all life. In simpler organisms like bacteria and yeast, a single cell performs all necessary life functions. Moulds and higher plants and animals are made of many cells working together. Plant, fungal, and bacterial cells may have a cell wall, but only plant cells have chloroplasts for photosynthesis. Bacterial cells have a region called a nucleoid instead of a true nucleus. Modern electron microscopes can magnify even more details inside cells.

Summary Points

- Microorganisms are invisible to the unaided eye but found everywhere.
- Living organisms can be unicellular (bacteria, protozoa) or multicellular (plants, animals, some fungi and algae).
- Cells have cell membrane, cytoplasm, and usually a nucleus. Plant, fungal, and bacterial cells have a cell wall.
- The shape and size of a cell relate to its function.
- Microbes help in recycling waste, maintaining soil fertility, and in preparation of many foods.
- Some microbes can cause disease, while many are useful in daily life.
- Microalgae are important for oxygen balance and food chain support.